

Software Test Plan (STP) - Railway Reservation System

Project: Railway Reservation System

Team No: Team 1 (Jackfruit Problem), Section 5C

Version: 1.0

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1. Introduction

Purpose: This document defines the test plan for the Railway Reservation System v1.0. It outlines objectives, scope, strategy, resources, schedule, and responsibilities for testing.

Scope: Testing covers all 22 functional requirements (RRS-F-001 to RRS-F-022), 5 non-functional requirements (RRS-NF-001 to RRS-NF-005) and 5 security requirements (RRS-SR-001 to RRS-SR-005) defined in SRS v1.0: registration and authentication, train search, booking, payment and ticketing, cancellation and refund, booking history, and administration/reporting. Actual payment-gateway settlement and physical railway signalling systems are excluded, as they are external interfaces per SRS 1.2.

References: SRS v1.0 — Railway Reservation System (Team 1, 05-09-2026), including the RTM in Section 8; SAD v1.0; C/C++ coding standards.

Definitions: RRS (Railway Reservation System), PNR (Passenger Name Record), FR (Functional Requirement), NFR (Non-Functional Requirement), SRS (Software Requirements Specification), RTM (Requirements Traceability Matrix), AC (Acceptance Criteria).

2. Test Items

- AuthModule — registration, login, account logout
- SearchModule + SeatInventory — train search and seat availability
- BookingModule — booking, berth preference, PNR generation, booking history
- PaymentModule — fare calculation, payment mode, ticket generation
- CancelModule — cancellation, seat restoration, refund calculation and status
- AdminModule — schedule management, seat inventory, occupancy/revenue reports
- File/DB Storage + AuditLog — persistence and audit trail

3. Features to be Tested

Every requirement below is taken from SRS v1.0, together with the test case ID named in its acceptance criteria.

Req ID	Feature	Test case(s)
RRS-F-001	Passenger registration	TC-Auth-01
RRS-F-002	Passenger authentication	TC-Auth-02
RRS-F-003	Lockout after 3 failed logins	TC-Auth-03
RRS-F-004	Train search by source/destination/date	TC-Search-01

Req ID	Feature	Test case(s)
RRS-F-005	Per-class seat availability display	TC-Search-02
RRS-F-006	Alternative train list when full	TC-Search-03
RRS-F-007	Book ticket (train, class, passengers)	TC-Book-01
RRS-F-008	Seat/berth preference	TC-Book-02
RRS-F-009	Unique PNR generation	TC-Book-03
RRS-F-010	Seat count decrement on booking	TC-Book-04
RRS-F-011	Reject booking beyond inventory	TC-Book-05
RRS-F-012	Fare calculation	TC-Pay-01
RRS-F-013	Payment mode recorded	TC-Pay-02
RRS-F-014	Ticket generation with all fields	TC-Pay-03
RRS-F-015	Cancel ticket by PNR	TC-Cancel-01
RRS-F-016	Restore seats on cancellation	TC-Cancel-02
RRS-F-017	Refund amount per policy	TC-Cancel-03
RRS-F-018	Refund/cancellation status by PNR	TC-Cancel-04
RRS-F-019	View booking history	TC-Admin-01
RRS-F-020	Add/update/remove train schedule	TC-Admin-02
RRS-F-021	Configure seat inventory	TC-Admin-03
RRS-F-022	Occupancy/revenue report	TC-Admin-04
RRS-NF-001	Search response ≤ 2 seconds	TC-Perf-01
RRS-NF-002	No over-allocation of seats	TC-Rel-01
RRS-NF-003	Records persist across restart	TC-Rel-02
RRS-NF-004	Usable without documentation	TC-UX-01
RRS-NF-005	Builds/runs on Windows and Linux	TC-Port-01
RRS-SR-001	Password masked and hashed	TC-Sec-01
RRS-SR-002	Admin functions restricted	TC-Sec-02
RRS-SR-003	Input validation / overflow prevention	TC-Sec-03
RRS-SR-004	Timestamped audit log	TC-Sec-04
RRS-SR-005	Cancelled/invalid PNR not reusable	TC-Sec-05

4. Features Not to be Tested

- Actual payment-gateway settlement — payment is simulated per RRS-F-013
- Physical railway signalling and train-operations systems (external interfaces, SRS 1.2)
- Network/client-server multi-user access — listed as optional future scope in SRS 3.4

5. Test Approach / Strategy

Levels:

- Unit tests (module-level: bookTicket, cancelTicket, generatePNR, refund calculation)
- Integration tests (Booking/Cancellation Manager ↔ File/DB storage layer)
- System tests (end-to-end CLI booking and cancellation flows)
- Acceptance tests (UAT) organised into the suites named in SRS Section 6: Registration & Authentication, Search, Booking, Payment & Ticketing, Cancellation & Refund, Administration & Reporting, and Security

Types:

- Functional testing (core booking/cancellation features)
- Regression testing
- Performance testing — search response time measured against the 2-second target (TC-Perf-01)
- Boundary / edge-case testing (seat limits, invalid ages, invalid PNR formats)
- Portability testing — clean build and run on both Windows and Linux (TC-Port-01)

- Concurrency/stress testing of the lockSeat()/confirmSeat() protocol defined in SAD Sections 3.5 and 3.7 (TC-Rel-01)
- Persistence testing — records verified present after program restart (TC-Rel-02)
- Failure-path testing (payment declined after seat lock → seat must be released, not left locked)

Entry Criteria: Stable build delivered; train master data, fare master data and the cancellation-policy table available; test environment ready on both Windows and Linux.

Exit Criteria: Per SRS Section 6: all high-priority functional requirements implemented and verified, no critical NFR failures, and the RTM shows all test cases passed. In addition, 100% of planned test cases executed and 0 critical defects open.

5.1 Security Validation

Mapped to the two security objectives and five security requirements in SRS Section 5.1.

- Password entry is masked and the stored value is hashed, never plain text (RRS-SR-001, TC-Sec-01)
- A non-admin session cannot reach schedule management, inventory configuration, or reports (RRS-SR-002, TC-Sec-02)
- Invalid and out-of-range inputs — menu choices, dates, seat counts, over-long strings — are rejected without buffer overflow (RRS-SR-003, TC-Sec-03)
- Every booking, cancellation and admin action produces a timestamped audit-log entry (RRS-SR-004, TC-Sec-04)
- Actions on a cancelled or non-existent PNR are rejected and cannot modify state (RRS-SR-005, TC-Sec-05)
- Negative test: account lockout holds after 3 consecutive failed logins (RRS-F-003, TC-Auth-03)

6. Test Environment

Hardware: Standard PC with keyboard and monitor; local disk for persisting train, seat and booking records (SRS 3.2). No specialised hardware required.

Software: Railway Reservation System build compiled with GCC/G++, tested on both Windows and Linux to satisfy RRS-NF-005.

Tools: GDB (debugging), Valgrind (memory-leak detection), Unity/CUnit (C unit-testing framework), Jira/Excel (defect tracking).

Test Data: Train master data, fare master data, cancellation-policy table, dummy passenger accounts (valid and duplicate contacts), and a set of booked PNRs including already-cancelled ones for RRS-SR-005.

7. Test Schedule

Milestones:

- Test case design: 20-Sep-2026
- Environment setup: 22-Sep-2026
- Test execution start: 23-Sep-2026
- Test execution end: 30-Sep-2026
- UAT: 01-Oct-2026 to 03-Oct-2026

8. Test Deliverables

- Test Plan (this document)
- Test Cases (manual & automated)

- Test Scripts
- Test Data
- Test Execution Logs
- Defect Reports
- Test Summary Report

9. Roles and Responsibilities

Role	Name	Responsibility
Developer/Test Lead	Farheen Akhtar (PES2UG24CS164)	Prepare plan, coordinate execution
Developer	HANUMANTHA L (PES2UG24CS179)	Design & execute test cases, log defects
Developer	GIRISH JAYAN SENTHILKUMAR (PES2UG24CS170)	Support defect fixes and triage
Developer	MEENAKSHI P (PES2UG25CS811)	Approve test results, sign-off readiness

10. Risks and Mitigation

Risk	Mitigation
Double-booking of the same seat under concurrent access	Implement file/record locking around seat allocation and test extensively under concurrent load
Reservation data file corruption on crash	Use backup/temp-file write strategy; test crash-recovery scenarios
Incomplete test coverage due to limited team size	Prioritize critical booking and cancellation flows first

11. Assumptions & Dependencies

- Sample train-schedule data will be provided before test execution
- Compiler/toolchain (GCC/G++) is available and consistent across environments
- No real banking integration is required for this academic/sample project

12. Suspension & Resumption Criteria

Suspend testing if:

- Build crashes on more than 30% of planned test cases
- Test environment unavailable for more than 4 hours

Resume testing if:

- Blocking defects are resolved
- Build/environment is stabilized

13. Test Case Management & Traceability

This plan reuses the requirement IDs and test case IDs defined in SRS v1.0, so the RTM in SRS Section 8 extends directly from this document with no re-mapping. The full requirement-to-test-case mapping is given in Section 3 above; the SAD adds the design module for each requirement in its Section 3.8.

Traceability chain: SRS requirement (RRS-F / RRS-NF / RRS-SR) → SAD design module (DS-xxx) → STP test case (TC-xxx) → execution result recorded in the RTM Status column (N/P/A).

- Example: RRS-F-009 (unique PNR) → DS-Book-02 / BookingModule → TC-Book-03
- Example: RRS-NF-002 (no over-allocation) → SeatInventory lock/confirm protocol → TC-Rel-01
- Example: RRS-SR-002 (admin restriction) → AuthModule + AdminModule → TC-Sec-02

14. Test Metrics & Reporting

Metrics collected:

- % test cases executed
- % passed / failed
- Defect density
- Defect aging
- Requirement coverage

Reports:

- Daily execution status
- Final Test Summary Report

15. Approvals

Role	Name	Signature / Date
QA Lead	Farheen Akhtar Girish Jayan Senthilkumar	
Dev	Hanumantha L Meenakshi P	
Course Instructor	Dr. Pradeep Kumar Kenny S	14-09-2026